

Given a string w and a symbol a , let $delete(a, w)$ be the string w with all instances of a removed. For example, $delete(z, \text{jazzy})$ is jay and $delete(1, 00101110)$ is 0000 .

(a) Give a recursive function to compute $delete$.

Solution: Let a be a symbol and w a string. (Hint: there are at least two cases based on the recursive definition of strings, but you may need more than two!)

$$delete(a, w) := \begin{cases} \text{if} \\ \text{if} \end{cases}$$



Collaborators and sources:

- Bob Smith helped me. He took this course at UM in Fall 2020. He suggested that I needed two separate recursive cases.
- I looked at the slides at https://chengjianglong.com/slides/DS_Lecture_22.pdf for another perspective on recursive structures.
- I used ChatGPT. Here is the full transcript of my conversation: <https://chatgpt.com/share/6970e55f-e594-800e-88e3-c5eaa8caf7ce>

(b) Prove that $\text{delete}(a, x \cdot y) = \text{delete}(a, x) \cdot \text{delete}(a, y)$ for all strings x, y and all symbols a .

Solution: Let x, y be arbitrary strings and a an arbitrary symbol.

Assume... (what's the inductive hypothesis?)

There are ?? cases to consider:

- If $w = \varepsilon$, then

$$\text{delete}(a, x \cdot y) = \text{delete}(a, \varepsilon \cdot y) \quad \text{because ?}$$

$$= \text{delete}(a, \varepsilon) \cdot \text{delete}(a, y) \quad \text{because ?}$$

$$= \text{delete}(a, x) \cdot \text{delete}(a, y) \quad \text{because } w = \varepsilon$$

- ...you fill in the rest of the cases

In all cases, we conclude that $\text{delete}(a, x \cdot y) = \text{delete}(a, x) \cdot \text{delete}(a, y)$. ■

Collaborators and sources:

- Same ChatGPT conversation as above. Here is the full transcript of my conversation:
<https://chatgpt.com/share/6970e55f-e594-800e-88e3-c5eaa8caf7ce>

- (c) Recall the function $\#(a, w)$ from problem session 1, which returns the number of occurrences of the symbol a in string w . For any string $w \in \{0, 1\}^*$, prove that $|\text{delete}(1, w)| = \#(0, w)$.

Solution: Let $w \in \{0, 1\}^*$.

In all cases, we conclude that $|\text{delete}(1, w)| = \#(0, w)$. ■

Collaborators and sources: None.